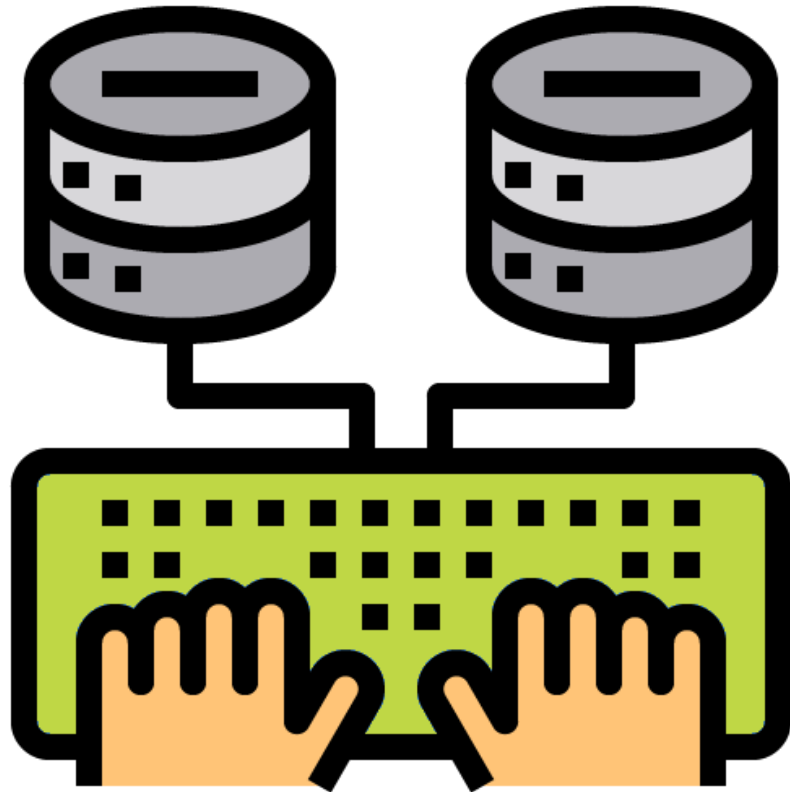




BIG DATA ANALYTICS

LECTURE 01

Overview of Big Data Analytics

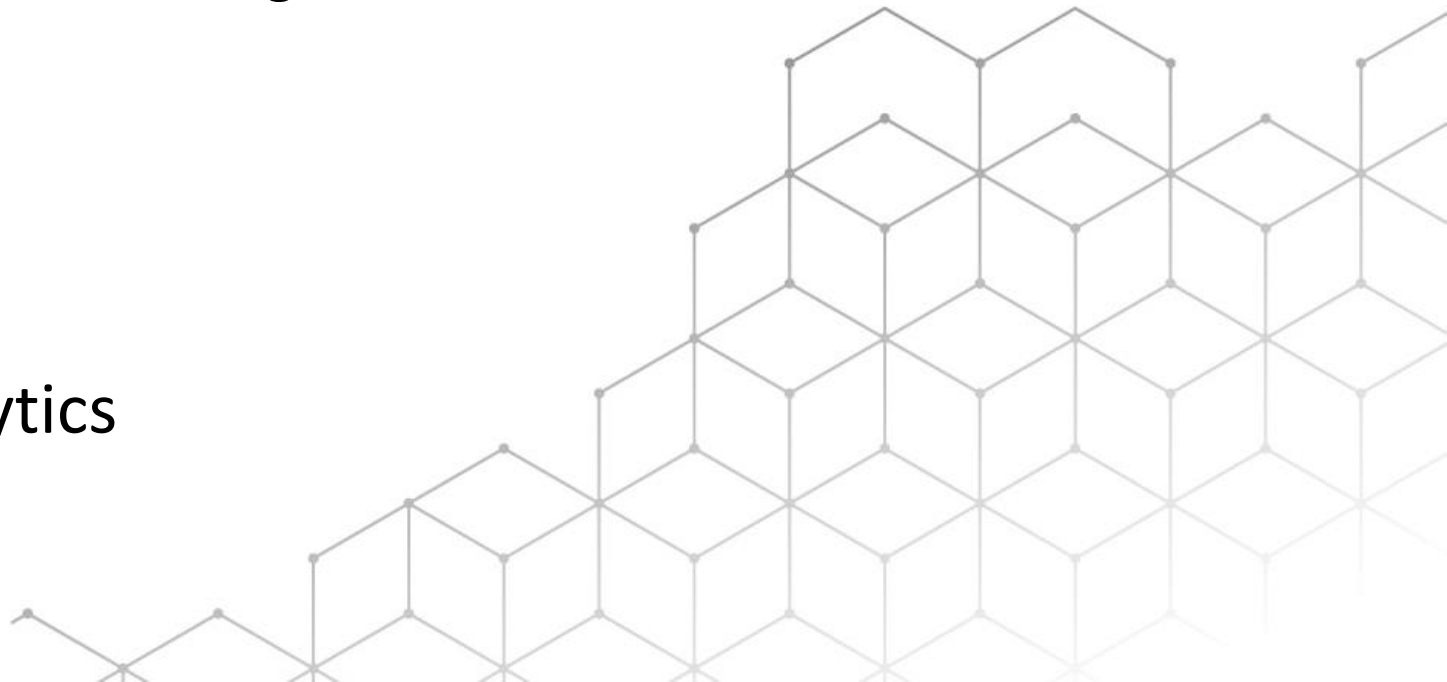
Lecturer:

Márcia Lourenço Baptista

m.baptista@novaims.unl.pt

Agenda for Today

- Course Overview:
 - Teaching Staff
 - Learning Objectives & Selling Points of the Course
 - Requirements
 - Course Objectives, Structure and Grading
 - Software and Tools
 - Bibliography
- Project Overview
- Understanding Big Data
- Importance of Big Data Analytics



NOVA

IMS

Information
Management
School

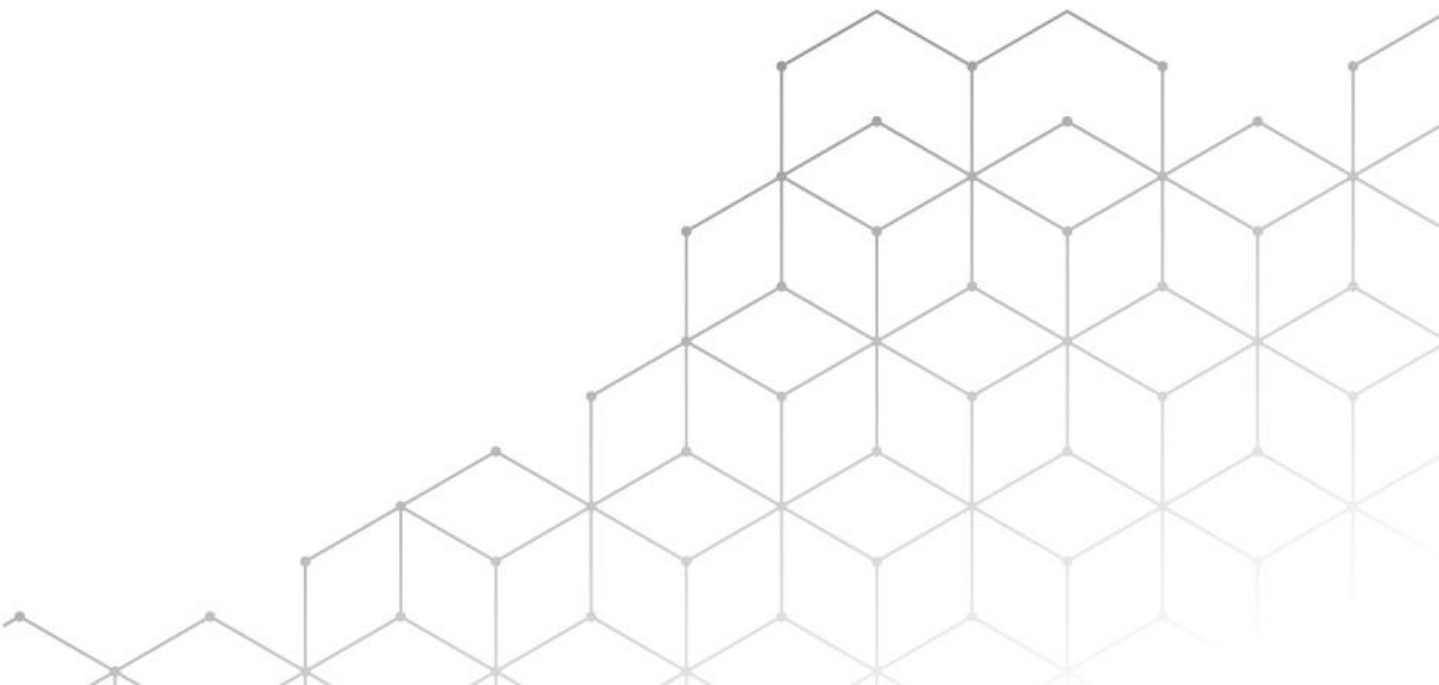
Teaching Staff

Lecturer:

Márcia L. Baptista

m.baptista@novaims.unl.pt

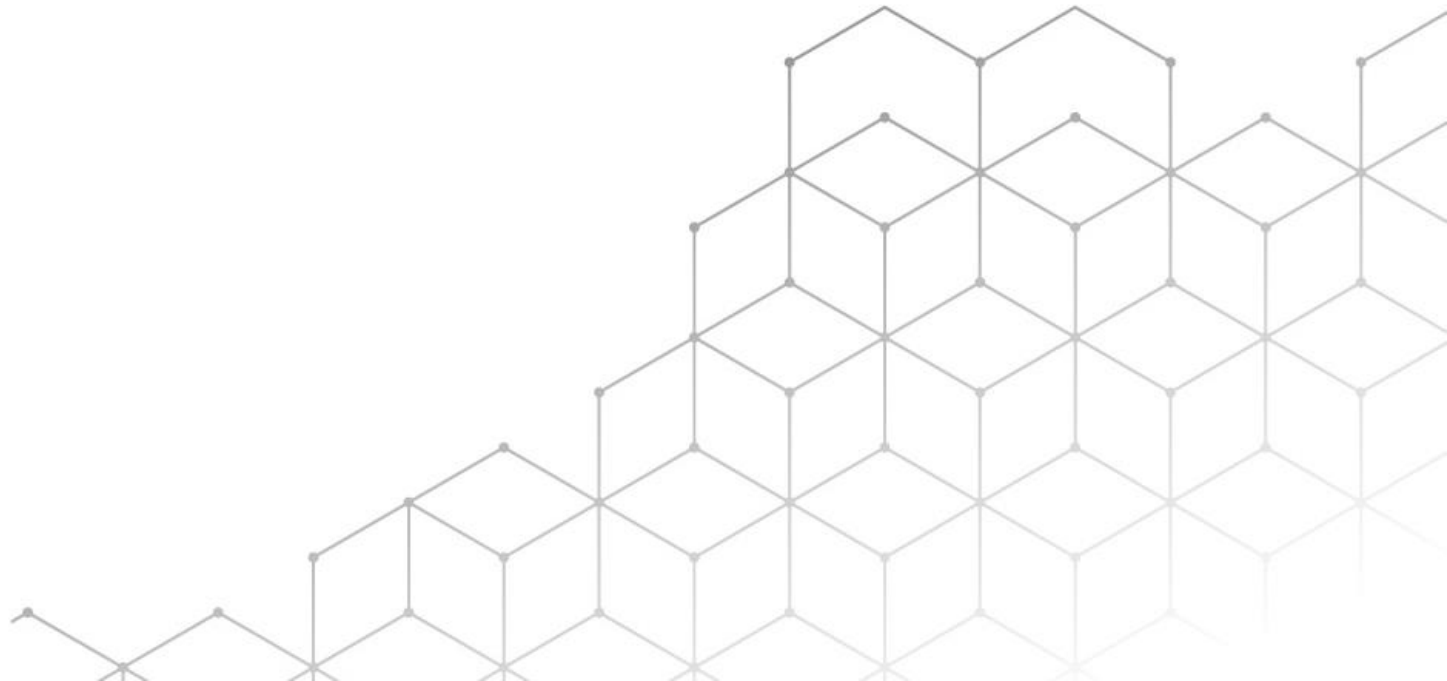
Office: 137 Colégio de campolide



Learning Objectives

- LO1 – Understand the Core Concepts and Ecosystem of Big Data
- LO2 – Develop Hands-on Skills with Big Data Technologies
- LO3 – Apply Advanced Analytics and Machine Learning to Big Data

OBJECTIVES

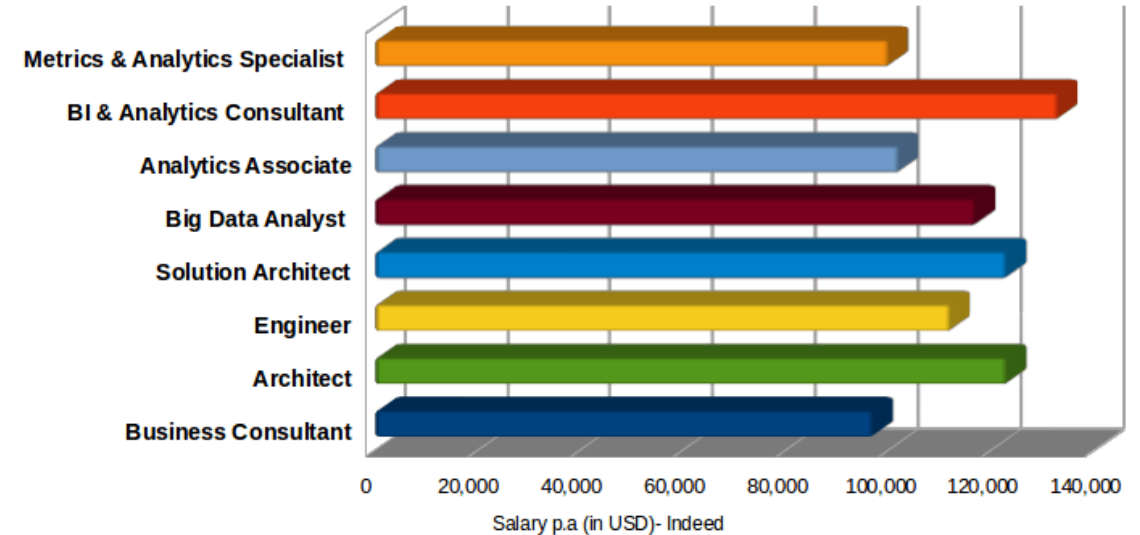


Selling Points

- ❖ Hands-On Practical Experience.
 - ❖ Industry-Relevant Curriculum.
 - ❖ Career-Boosting Skill Set.
-
- Advance your career in data science, data analytics, and big data engineering

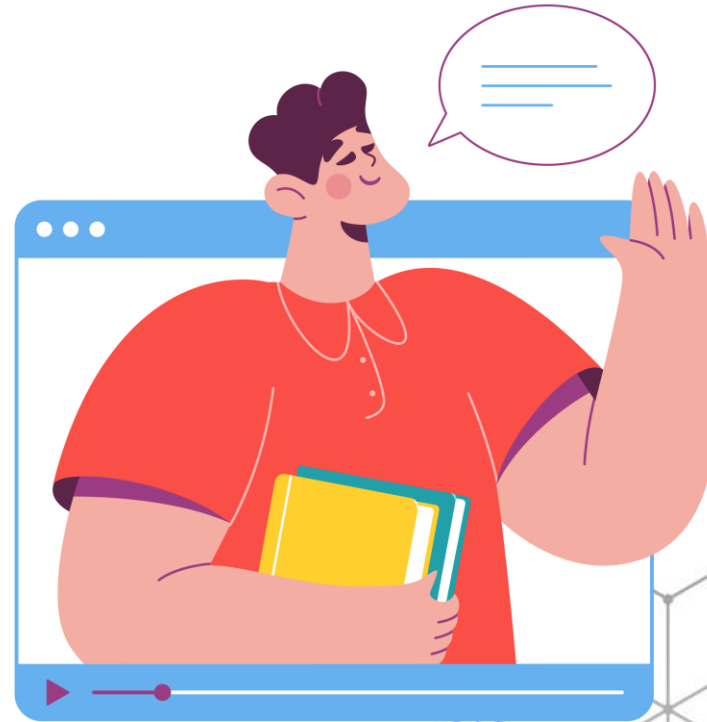
<https://www.itjobswatch.co.uk/jobs/uk/big%20data.do>

Big Data Analytics Job Titles & Salaries



Expectations

- During **Lectures**, you will learn what is data analytics, and why it matters.
- During **Labs**, you will develop hands-on experience with **Databricks** and Spark



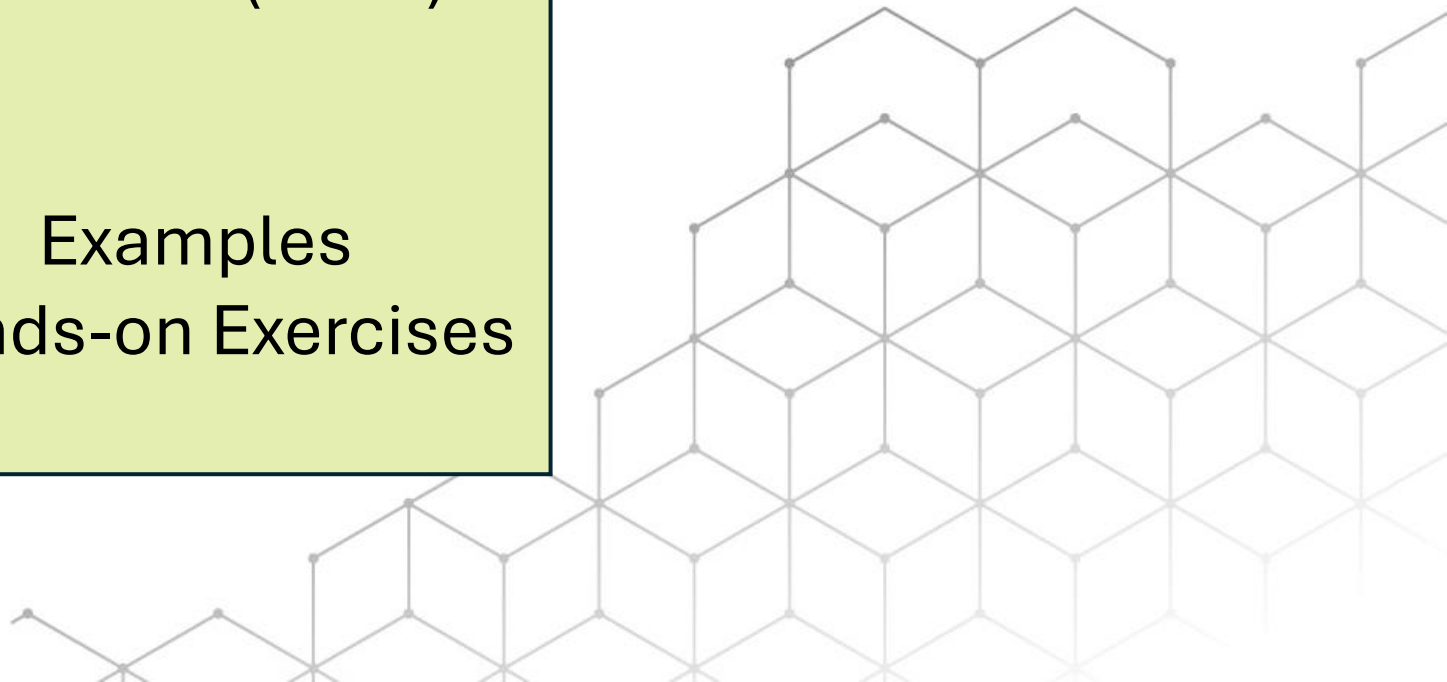
Course Structure

14 Theoretical
Lectures (**TP**)

Exposure to
Concepts
Practical Examples

14 Practical
Lectures (**Labs**)

Examples
Hands-on Exercises



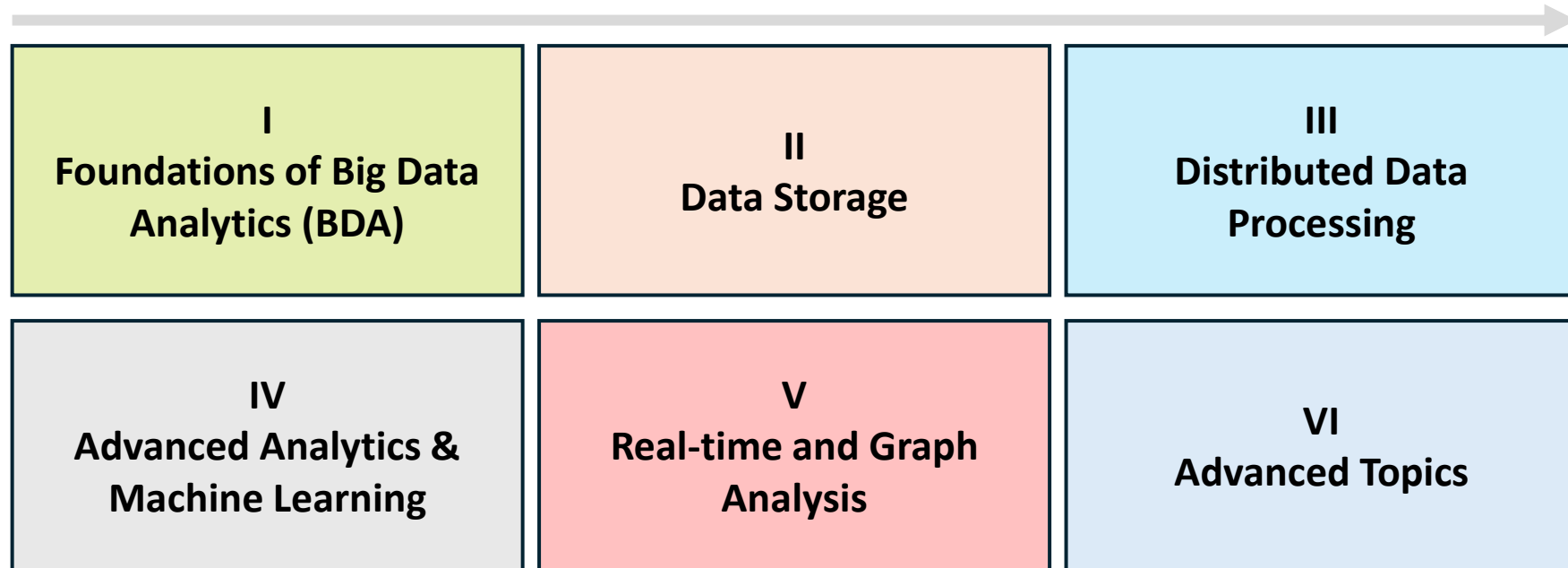
Theoretical Lectures

Theoretical Lectures (TP)

14 Lectures of 1.5 hours:

- I. Foundations of BDA
- II. Data Storage
- III. Distributed Data Processing
- IV. Advanced Analytics & ML
- V. Real-Time & Graph Analytics
- VI. Advanced Topics & Closing

Main Topics



Topics Description and Lectures

Main Areas	Topics Covered	Lectures
1. Foundations of BDA	Overview of Big Data Analytics, Distributed Processing, Big Data Platforms (Hadoop, Spark, Databricks)	TP01, TP02, TP03
2. Data Storage	Storage (HDFS, NoSQL, Data Lakes), Big Data SQL, RDDs, Query Optimization	TP04, TP05, TP06, TP07, TP08
3. Distributed Processing	Map Reduce	TP09
4. Analytics & ML	Spark MLlib, AutoML	TP10, TP11
5. Real-Time & Graph Analytics	Streaming (Kafka, Flink), GraphX	TP12, TP13
6. Advanced Topics	Advanced Topics	TP14

Lecture Map

TP Name	Practical Lecture
TP01: Overview of Big Data Analytics	Initial Practical Class
TP02: Distributed Computing	Centralized vs Distributed Computing
TP02: Big Data Platforms	Overview of Databricks Environment
	Creating visualizations and dashboards in Databricks
	Writing and executing a basic MapReduce job in Python
TP03: Distributed File System	Loading data into BDFS and running queries
TP04: NoSQL Databases	Running SQL queries in Databricks
TP05: Modern Data Storage Options	Delta Lake
TP06: RDDs Datasets	RDDs & Transformations
TP07: Query Optimization	Partitioning, caching & query optimization in Spark
TP08: Map Reduce	Map Reduce
TP09: Machine Learning in MLlib	Training ML models in MLlib & AutoML
TP10: AutoML and MLFlow for Big Data	Using MLflow to manage & deploy models
TP11: Streaming	Real-time data streaming using Spark Streaming
TP12: Graph Processing	Graph analytics with GraphX
TP14: Advanced Topics	Project Presentations

NOVA

IMS

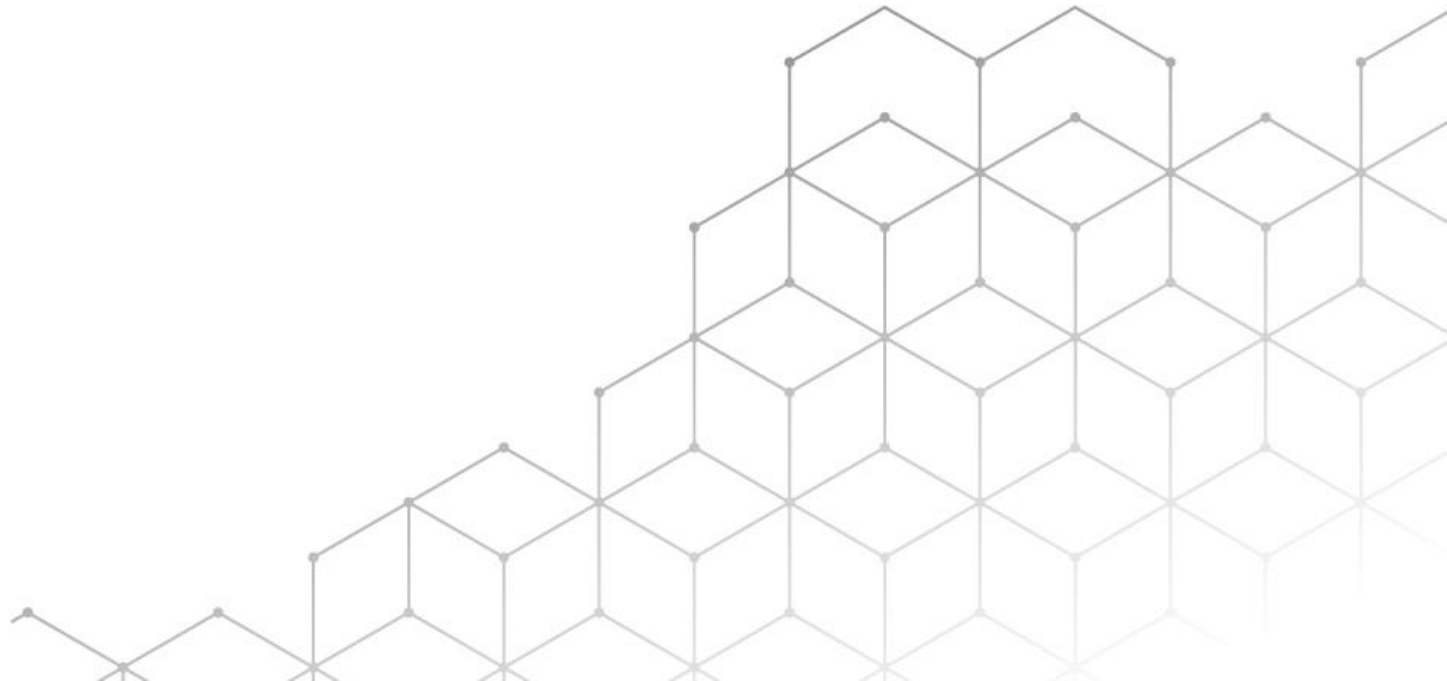
Information
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Labs

Practical
Lectures (**Labs**)

14 Lectures of 1.5 hours
Hands-on experience with

- Databricks



Grading

First
Option:

**Quizzes
(30%)**

**Project Assignment
(70%)**

Second
Option:

Exam (100%)

Grading: Quizzes

Quizzes (30%)

- Moodle 3 Quizzes in Class
- **No penalties** for wrong answers.
- 15 min at the end of third lecture onwards
- 5th Lecture, 9th Lecture, 12th Lecture
- End of lecture

Group Project

Group Assignments (70%)

- Group Activity (3 to up to 5 elements).
- Assignment released/submitted on Moodle
- **Penalties** will apply to late deliveries, max 10 days (-0.5 for each day).

Grading: Second Season Exam

**Exam
(100%)**

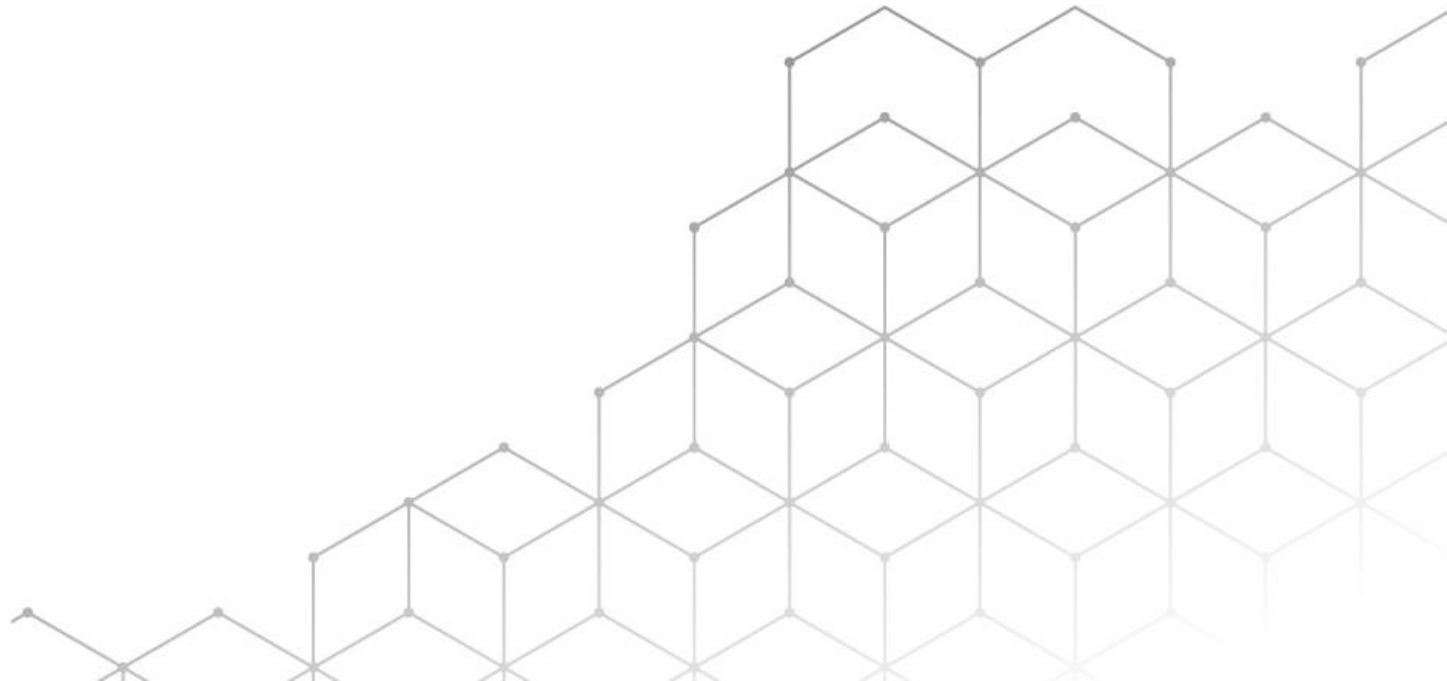
- Several multiple-choice questions
- One or two open ended questions
- No consultation
- Formulary provided in exam

Software & Tools

- All the practical tasks in this course will be done in the Databricks cloud environment; as such, the students will not be required to install any software.
- Register for a community account: <https://www.databricks.com/>

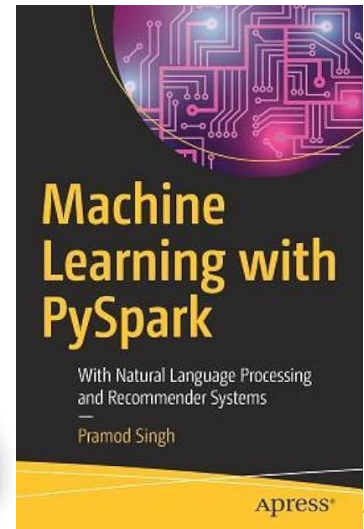
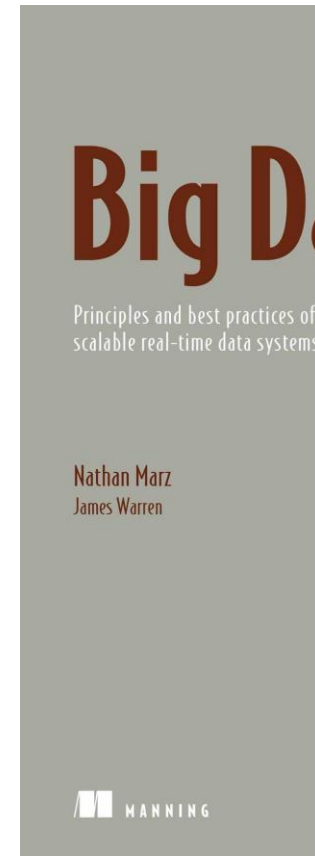


databricks®



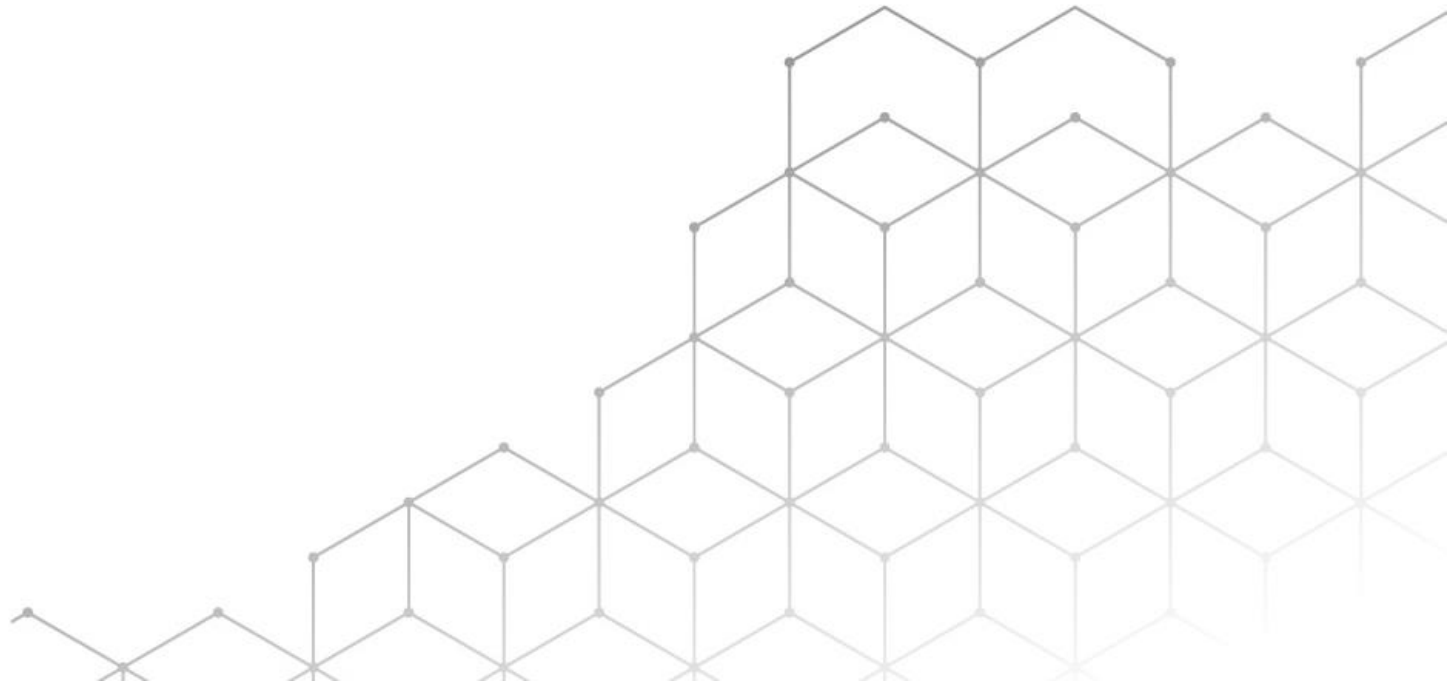
Bibliography

- Big Data: Principles and Best Practices
- Machine Learning with PySpark



Groups

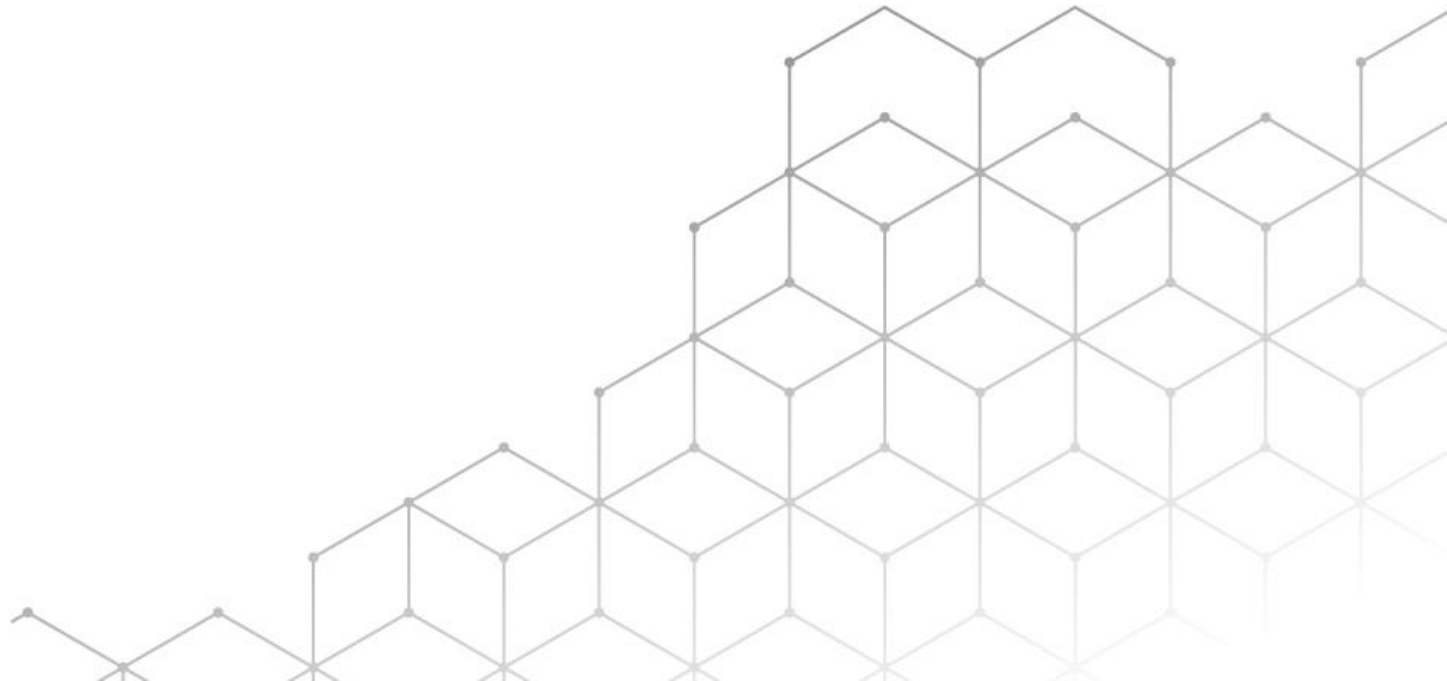
- Groups should have up to five members (minimum of 3 elements).
- Moodle
- February 28, 23:59 = Deadline for Groups



Important Dates

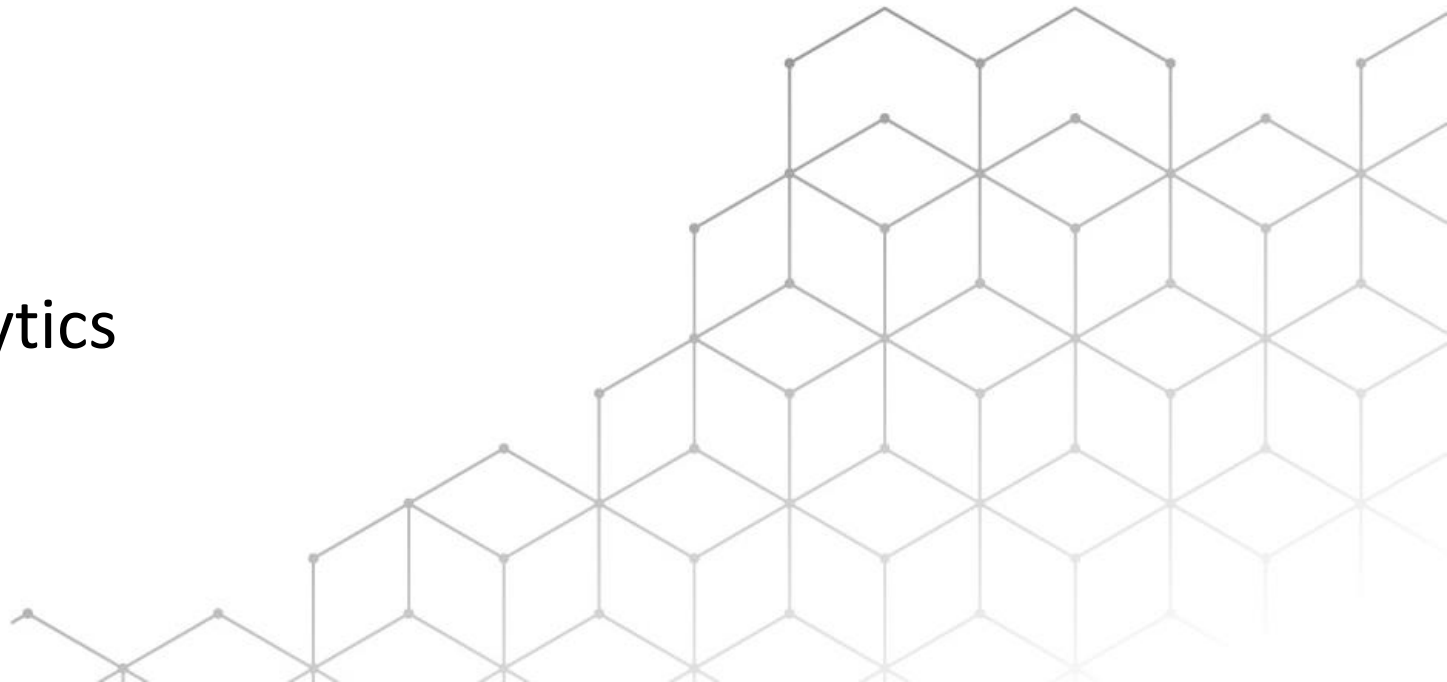


- Group Forming: 28/02/2025
- Project Delivery: 25/05/2025
- Resit Exam Date:
 - 16/07/2025 Time 2nd Season 15:30:00

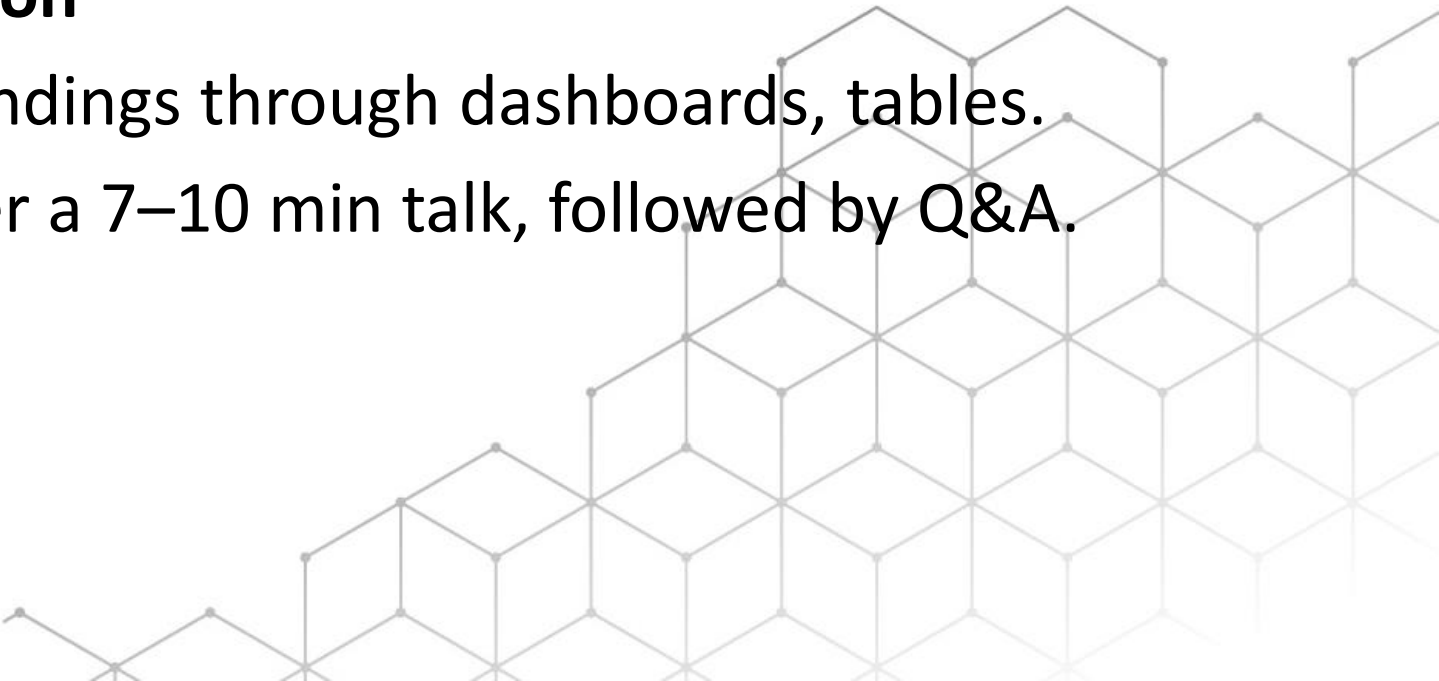


Agenda for Today

- Course Overview
- Project Overview:
 - Project Description
 - Possibilities
 - Deliverables
 - Gantt Chart & Planning
- Understanding Big Data
- Importance of Big Data Analytics



Project Description

1. **Problem Definition:** Identify a relevant Big Data problem.
 2. **Data Engineering:** Obtain, and preprocess datasets.
 3. **Data Science and Analysis:** machine learning, statistics, and BI.
 4. **Results and Data Visualization**
 5. **Visualize Results:** Present findings through dashboards, tables.
 6. **Project Presentation:** Deliver a 7–10 min talk, followed by Q&A.
- 

Possibilities

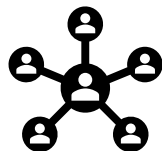
- 1. Business & Finance

- 
- Stock Market Prediction using Big Data & Spark Mllib
 - Fraud Detection using real-time transaction streams (Kafka & Spark Streaming)
 - Customer Segmentation using Clustering in Spark MLib

- 2. Healthcare & Environment

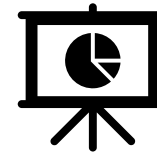
- 
- Disease Prediction & Analysis from healthcare datasets
 - Air Pollution & Climate Change Trends using Big Data visualization
 - Biodiversity Monitoring from satellite or sensor data

- 3. Social Media & E-commerce

- 
- Sentiment Analysis of Twitter/X Data using NLP in Spark
 - Recommendation System for e-commerce using Graph Analytics
 - Influence Analysis in social networks using GraphX

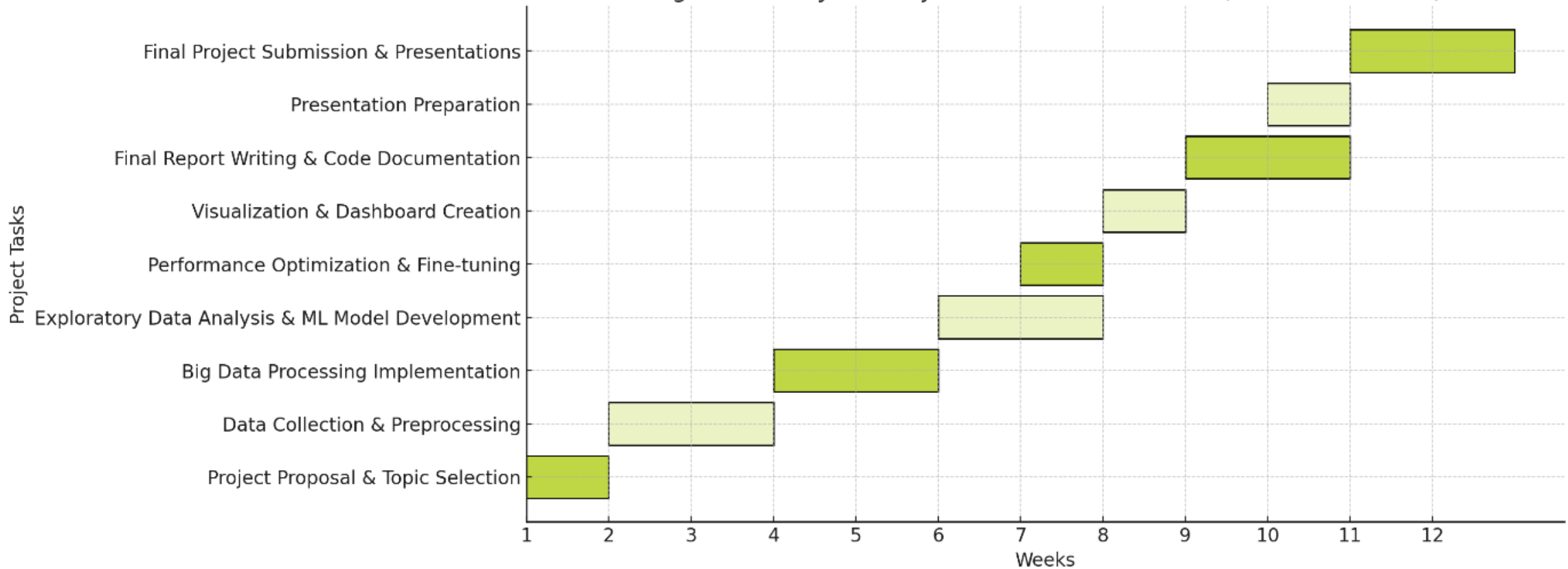
Deliverables

- Technical Report (3–5 pages **strict limit!**)
- Source Code (Databricks Notebook)
- Presentation (7–10 min, slides)



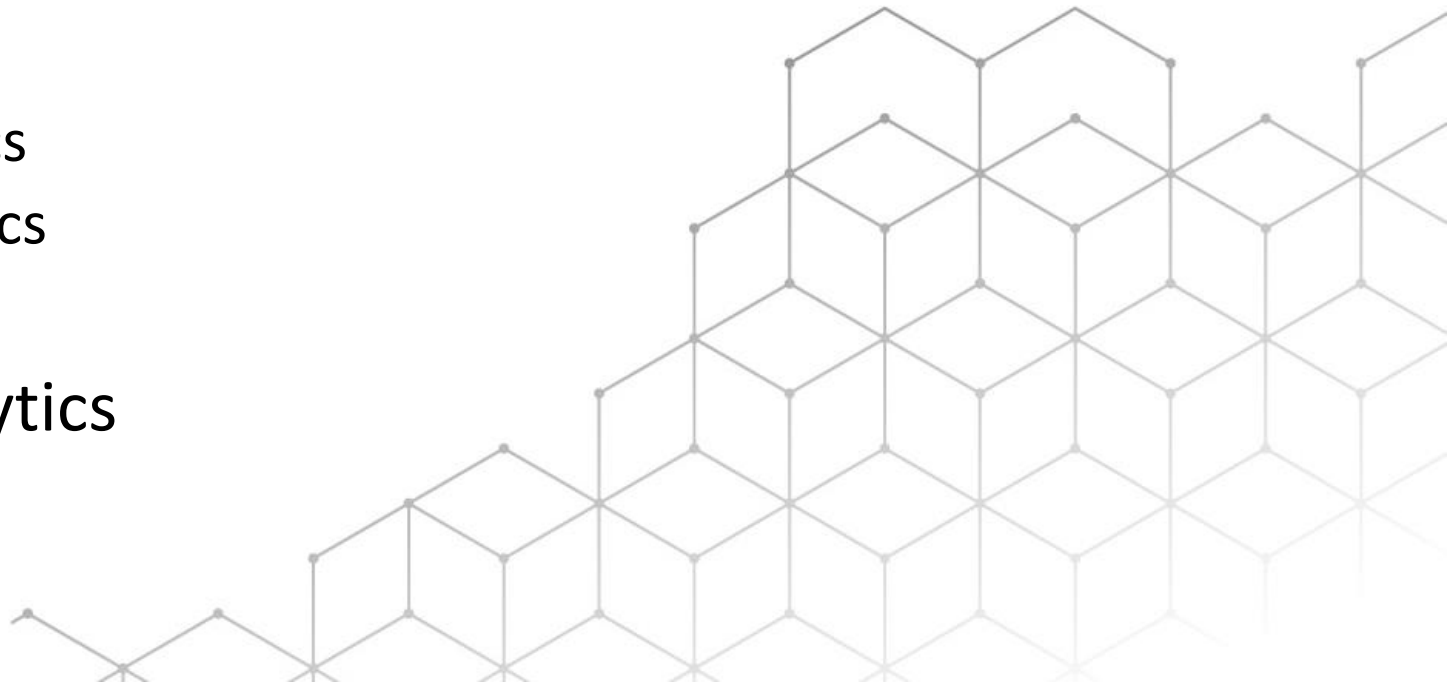
Gantt Chart and Planning

Big Data Analytics Project - 12 Week Timeline (Nova IMS Colors)



Agenda for Today

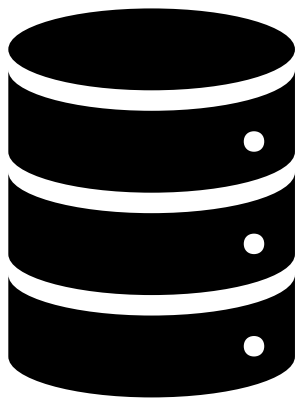
- Course Overview
- Project Overview
- Understanding Big Data:
 - Definition of Big Data
 - 5 Vs of Big Data
 - Definition of Big Data Analytics
 - Traditional vs Big Data Analytics
 - Three Types of Analytics
- Importance of Big Data Analytics



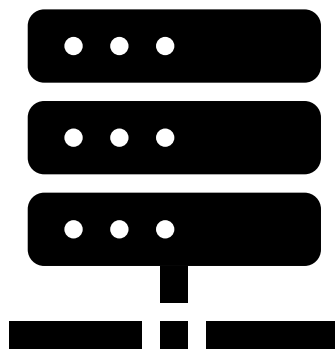
Definition of Big Data

- “Big data is high-volume, high-velocity and high-variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making.” – Gartner

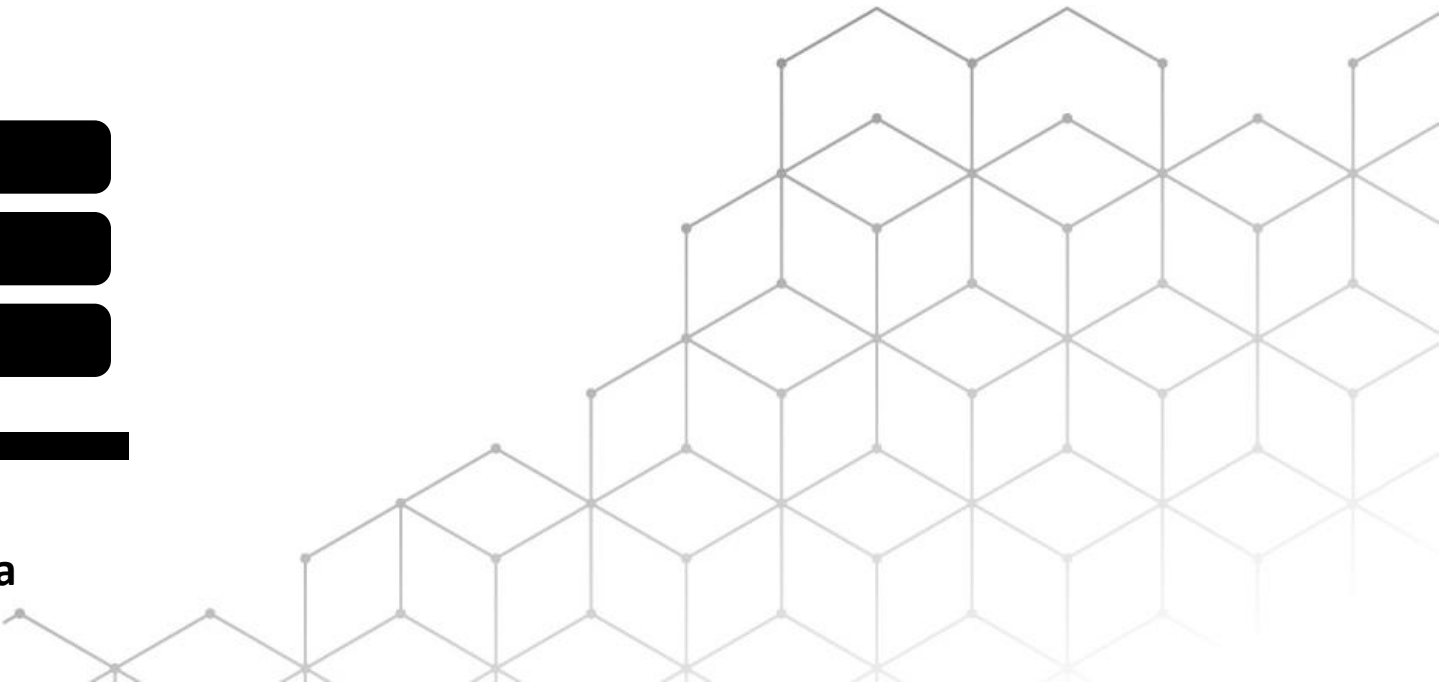
www.gartner.com/en/information-technology/glossary/big-data



Traditional
Data

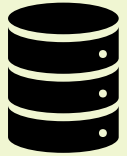


Big
Data



[Original] Properties of Big Data

Volume



Velocity



Variety

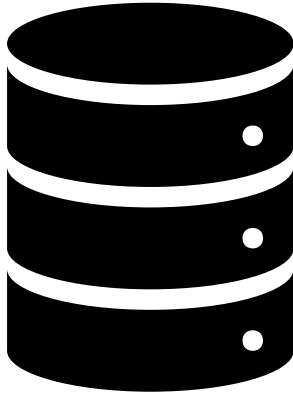


- **Volume:** High volume data
- **Velocity:** Data is produced at high speeds
- **Variety:** Data comes in many formats and sources

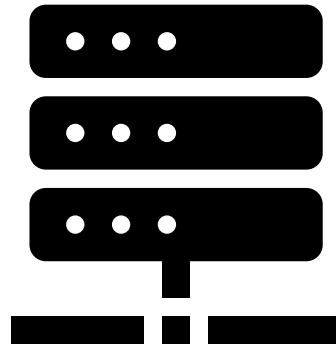
Volume: Scale of Data



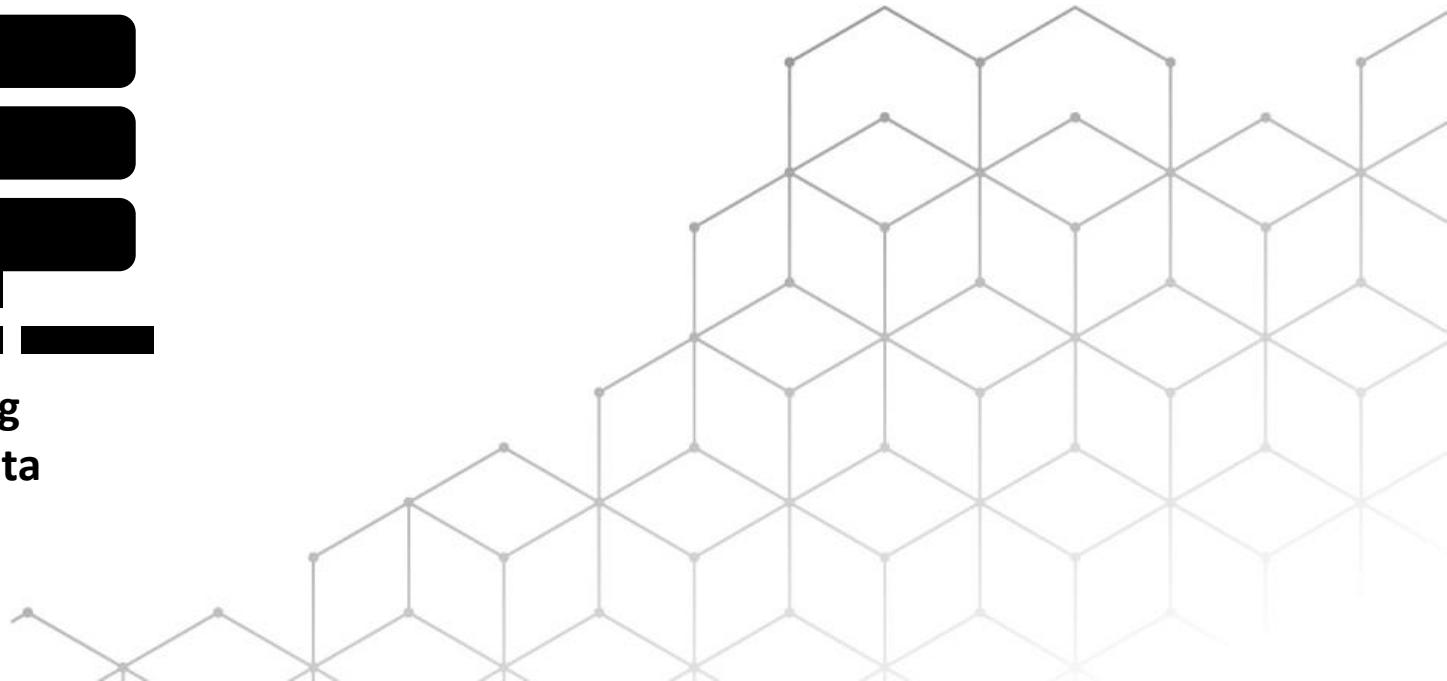
- Big data needs a whopping amount of memory space, typically **distributed between many** computers. Its size is measured in terabytes, petabytes, and even exabytes.



Traditional
Data



Big
Data



Velocity: Real-Time Processing

Velocity



- When working with big data, one's goal is to make extracting patterns from it as quick as possible.

But where do we encounter big data?

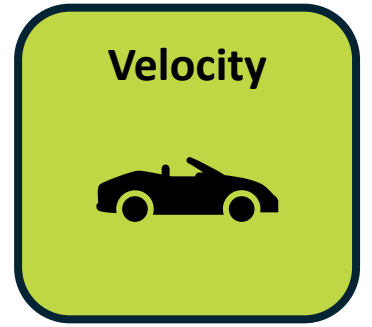
- The answer is: in increasingly more industries and companies.



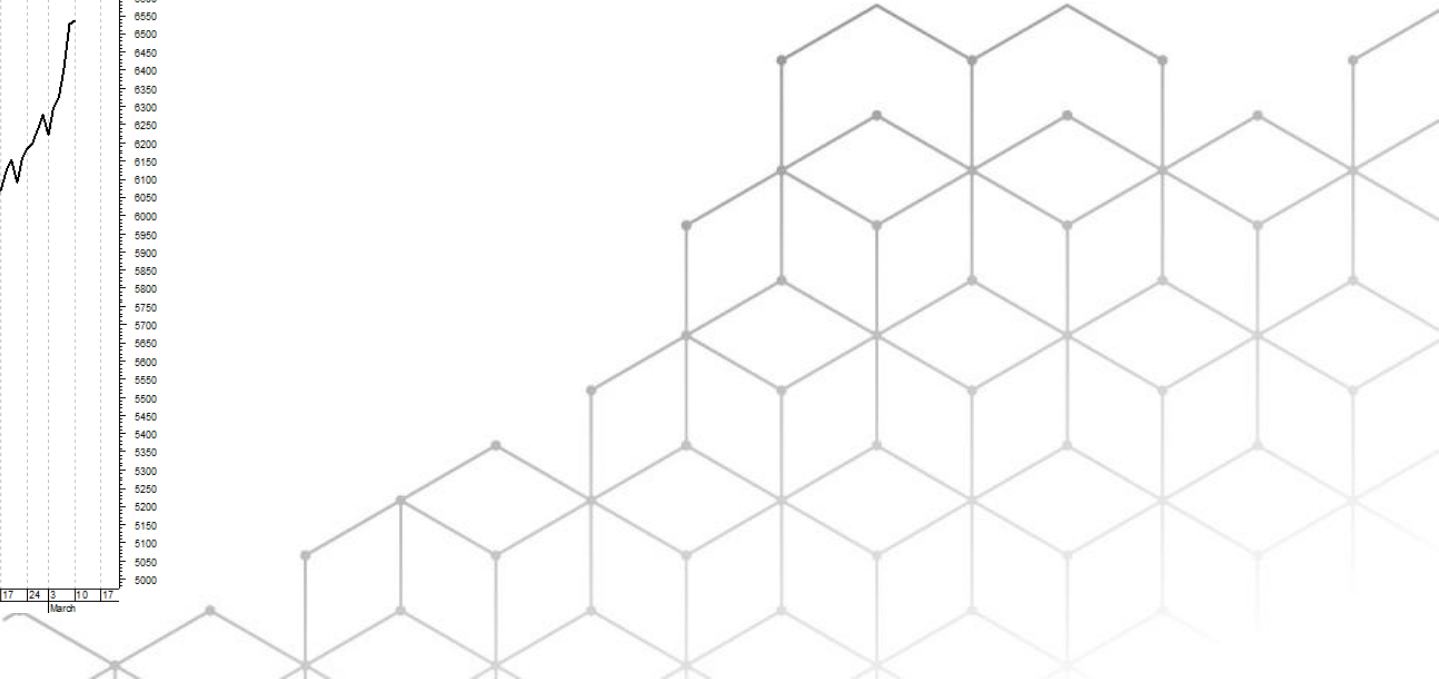
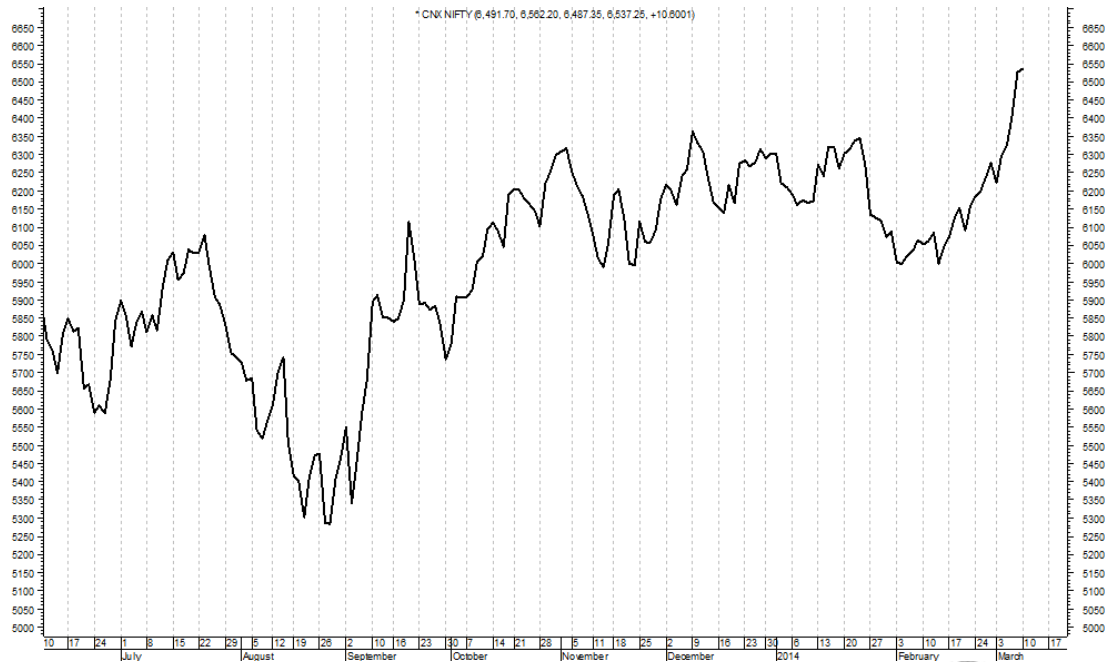
Smart Factory



Example of Data Velocity

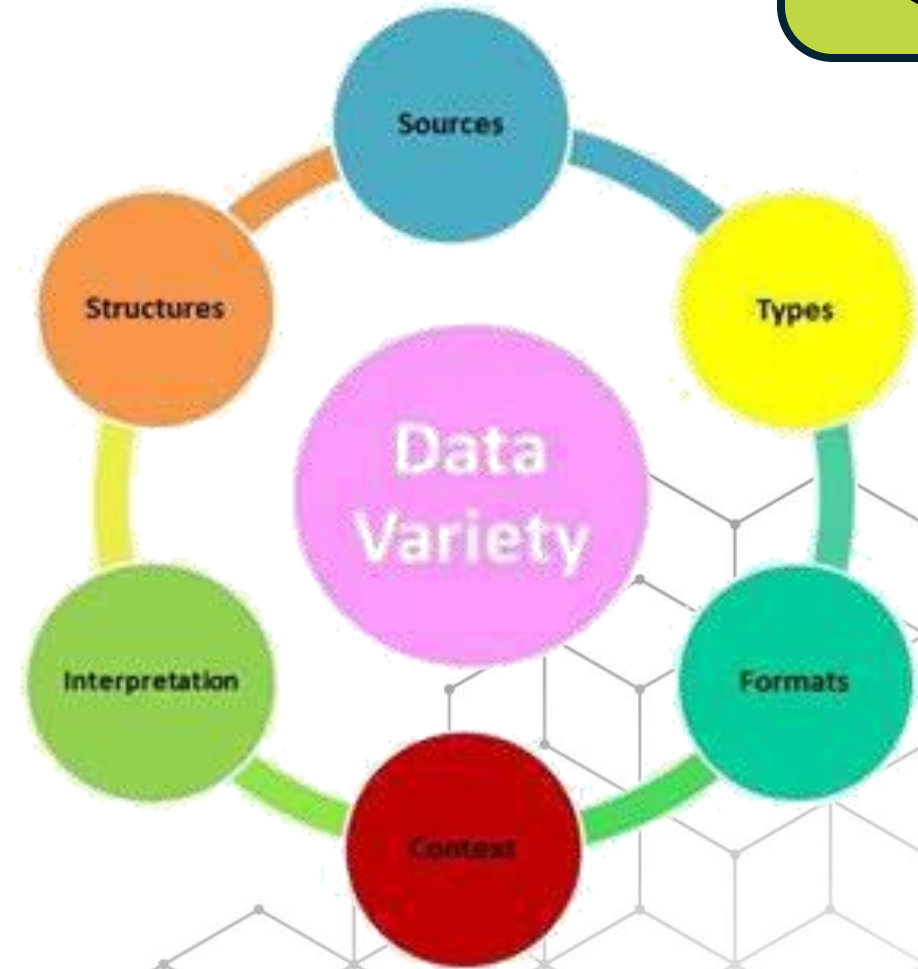


- Let's take 'financial trading data' for an extra example.
- What happens when we record the stock price at every 5 seconds? Or every single second?



Variety of Data

- Big data often implies dealing with images, audio files, mobile data, and others.
- It also means variety in **data sources**, data **format** and **type**, **context**, interpretation and **structures**.



Big Data in Short

- **Volume** — we need to process lots of data
- **Velocity** — we need to deal near-real time
- **Variety** — we need to handle structured, semi, and unstructured data from different sources
- **Note that** lots of data does not mean good data! We need to assess the quality of the data

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"After analyzing all your data, I think we can safely say that none of it is useful."

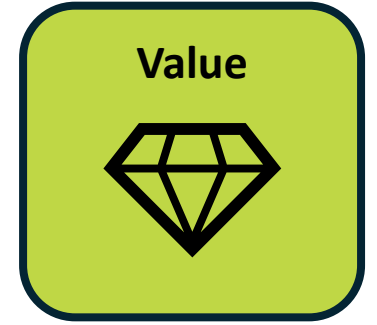
Veracity of Data



Veracity ensures that the data accurately represents the real-world scenarios it is intended to model.

- **High veracity** means the data is reliable, trustworthiness, and accurate.
- Focus in reducing uncertainty, errors, bias, and inconsistencies.
- **Example:** A satellite dataset with **sensor noise or missing values** has low veracity.

Value of Data



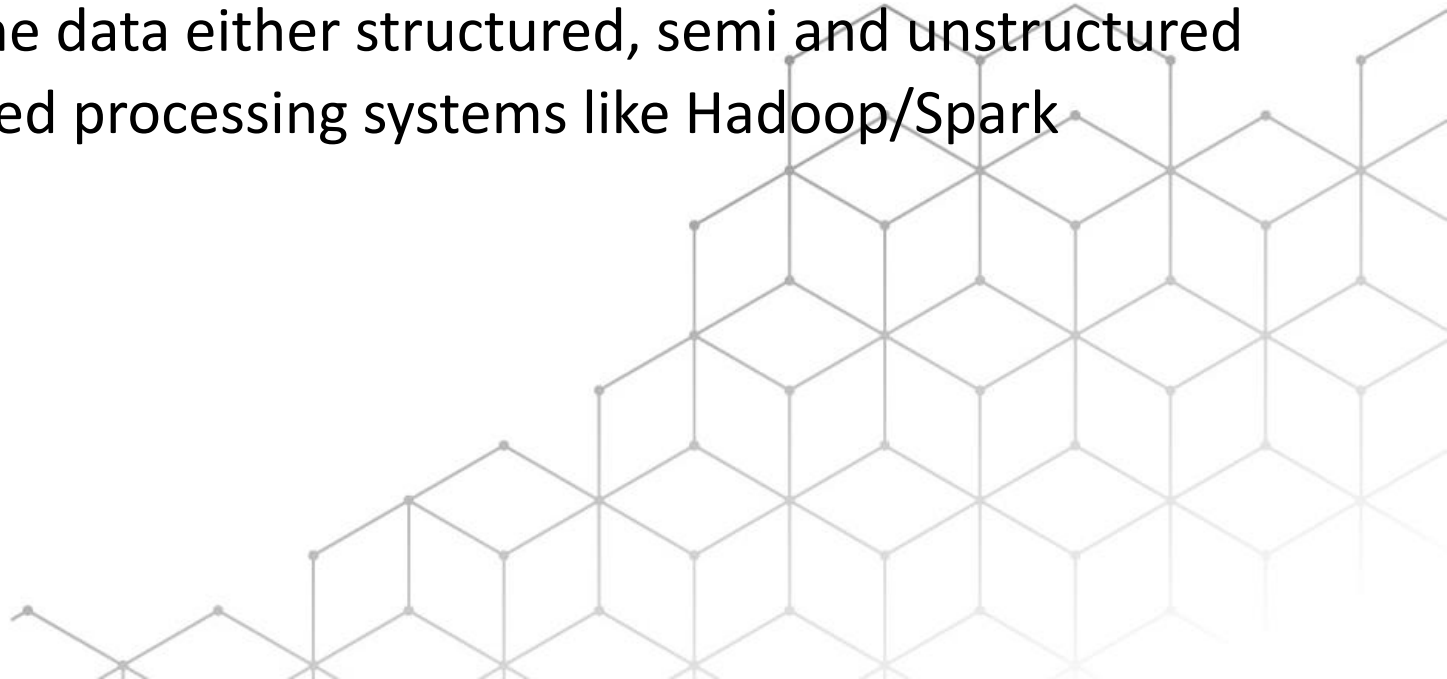
- The usefulness and impact of data for decision-making.
- Focus in on timeliness, accessibility, and business impact.
- **High Veracity, Low Value:** A perfectly clean dataset of ocean temperatures, but it only covers a tiny, irrelevant area.
- It is more difficult to have high value if veracity is low

Definition of Big Data Analytics

- **Big data analytics** refers to the systematic **processing and analysis** of large amounts of data and complex data sets, known as **big data**, to extract **valuable insights**.
- This process allows organizations to leverage data generated from diverse **sources**, including **internet-of-things (IoT) sensors**, **social media**, **financial transactions** and **smart device**.

Big vs Traditional Data Analytics

- Traditional analytics:
 - Type of Data: Conventional structured data (relational databases)
 - Processing: Query language (SQL)
- Big data analytics:
 - Type of Data: Big speed volume data either structured, semi and unstructured
 - Processing: Requires distributed processing systems like Hadoop/Spark

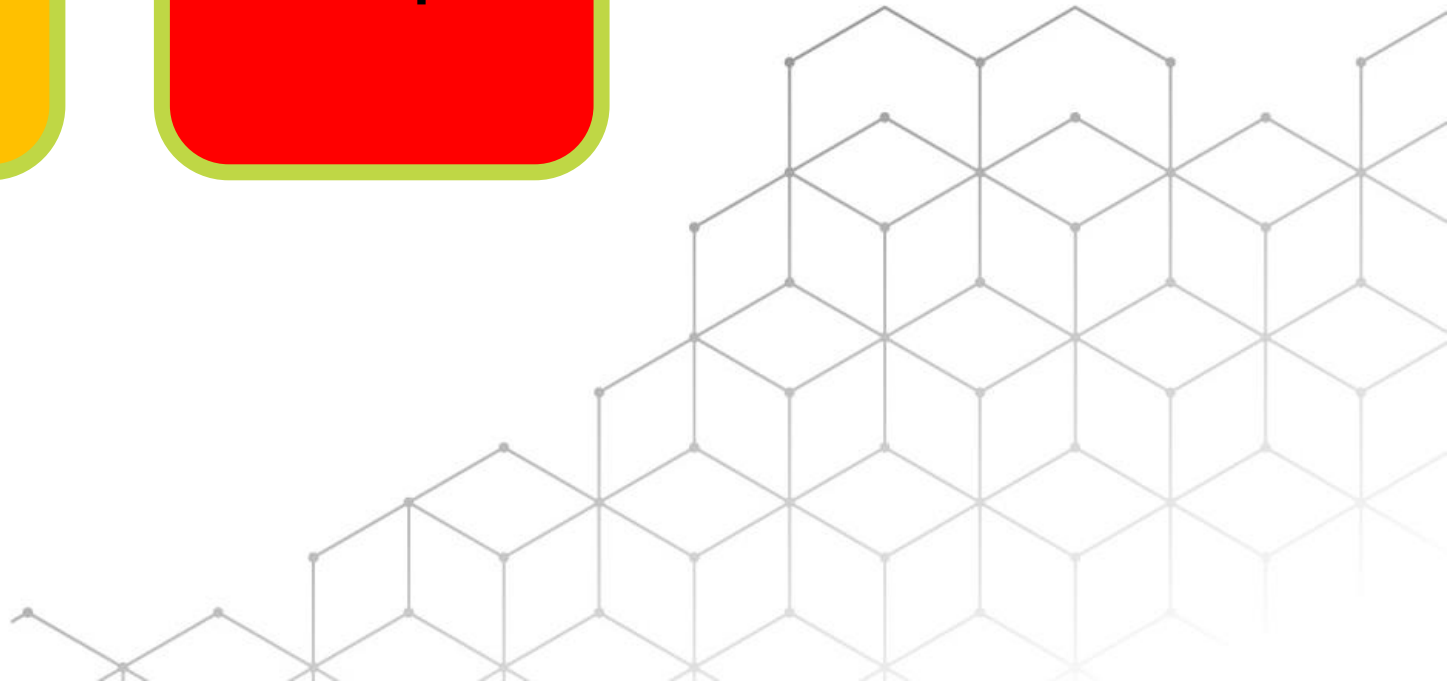


Evolution of Analytics

Descriptive

Predictive

Prescriptive

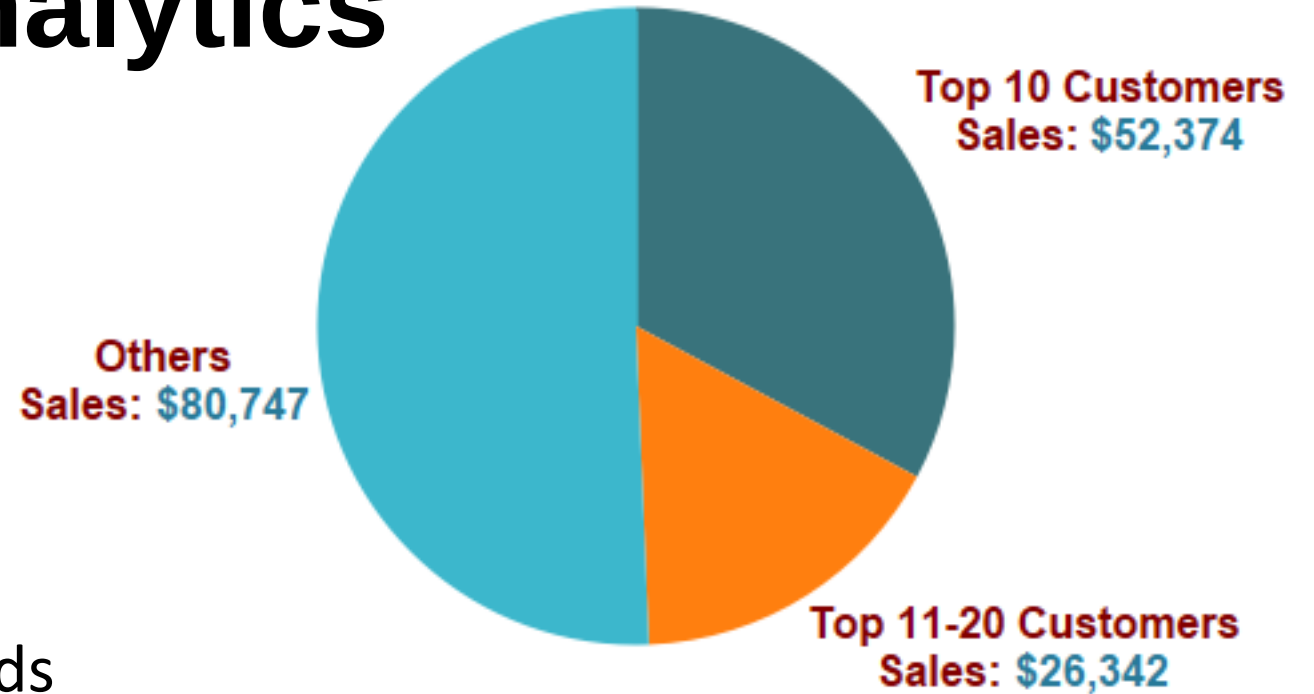


Descriptive Analytics

What happened **before**?

What is happening **now**?

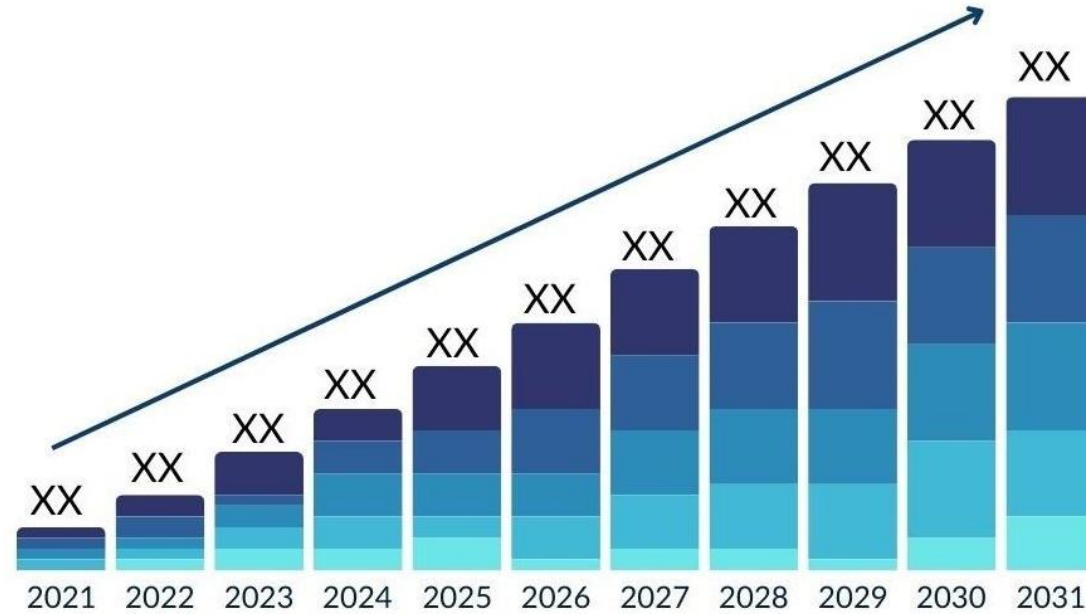
- Data warehouse
- Reports, queries, alerts, and trends
- Visualization
- Strategic and tacit knowledge



Predictive Analytics

What **will** happen?

- Forecasting
- Projections



Prescriptive Analytics

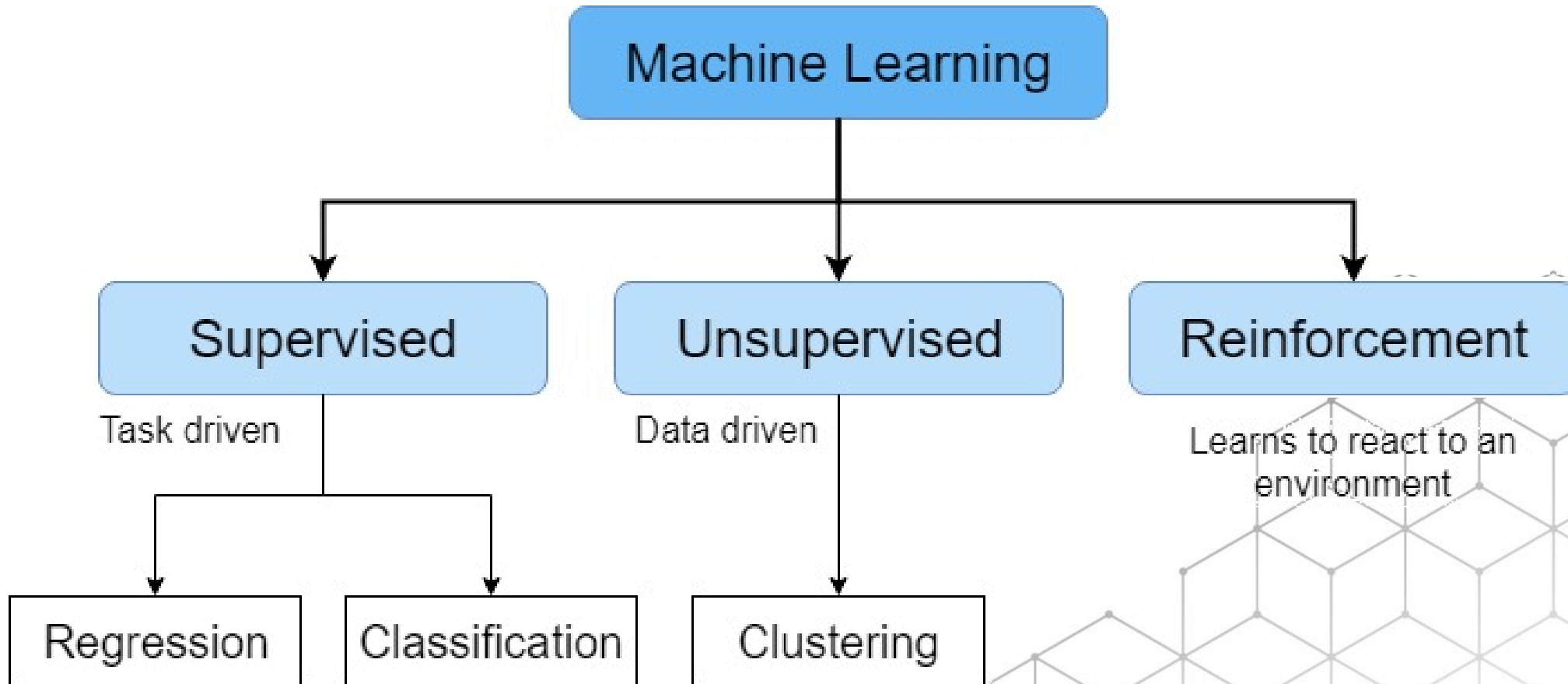
What should I do?

Why should I do?

- Machine Learning
- Explainable artificial intelligence
- Simulations
- Better decisions, more informed

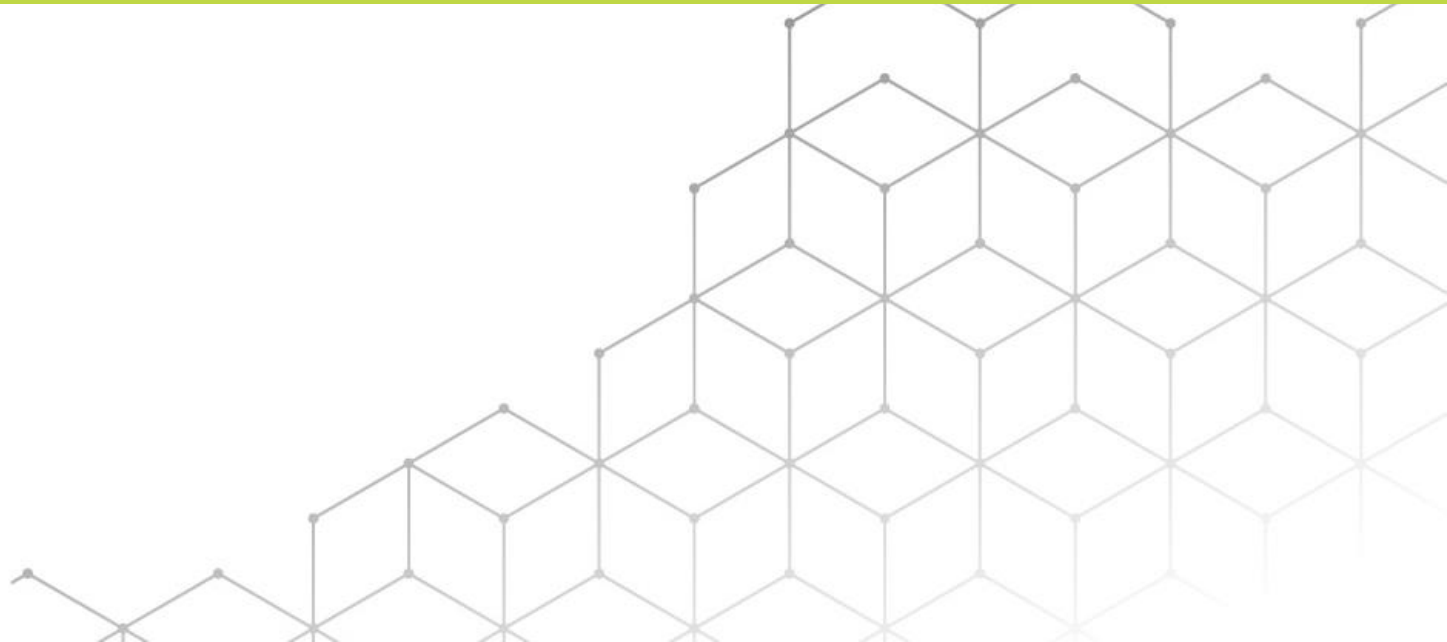


Machine Learning Models



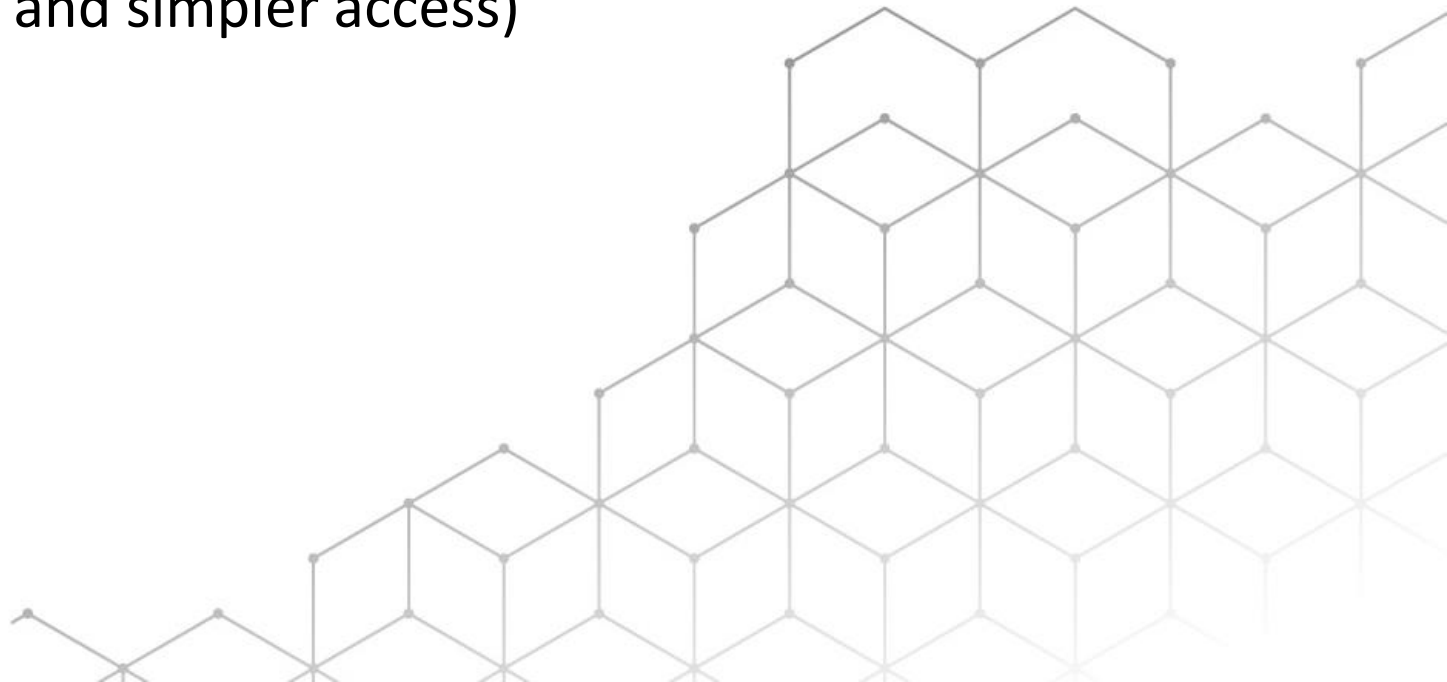
Agenda for Today

- Course Overview
- Project Overview
- Understanding Big Data
- Importance of Big Data Analytics
 - Why Big Data now?
 - Applications of Big Data
 - Warning



Why Big Data Now?

- Changes in the environments make big data analytics for all organizations:
 - More data than ever
 - Data management tools
 - Open source solutions (better and simpler access)
 - Artificial Intelligence



CERN and Physics

- This raw data is recorded onto servers at the CERN Data Centre at a rate around 1.5 CDs per second



Astronomy

- The Atacama Large Millimeter Array in Chile now anticipates adding 2TB of data to its archives every day



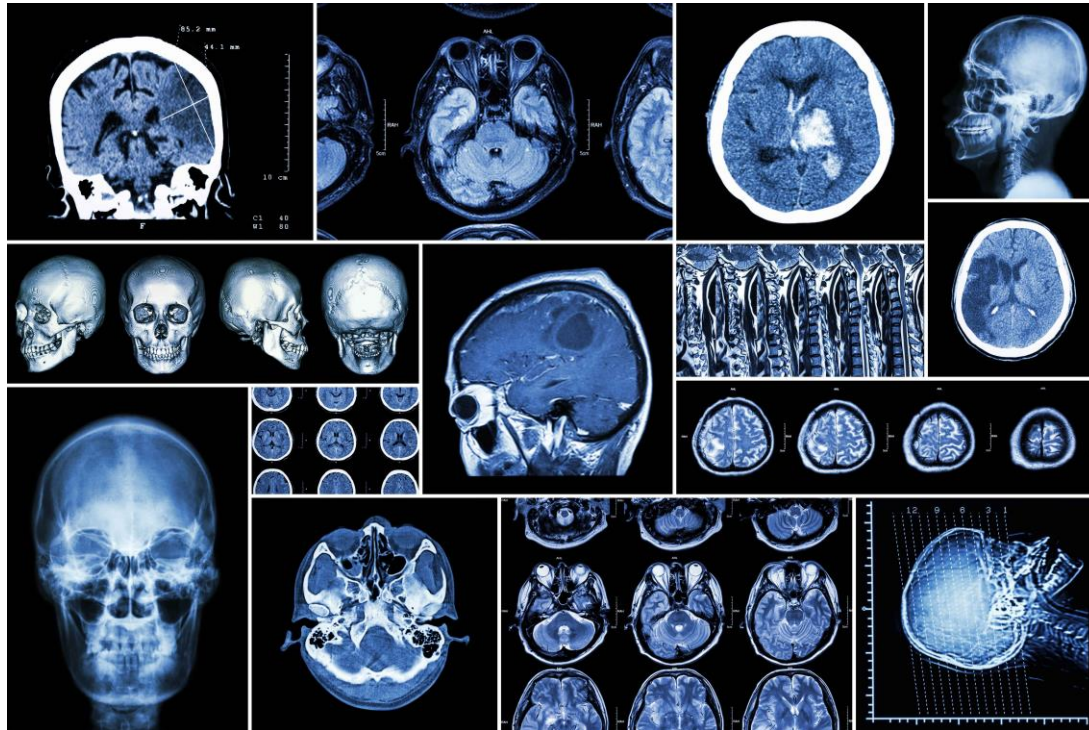
Human Genome

- Michael Schatz, an associate professor of quantitative biology says that 100 gigabytes is a good benchmark for projecting the storage requirements of any single human genome over the next decade



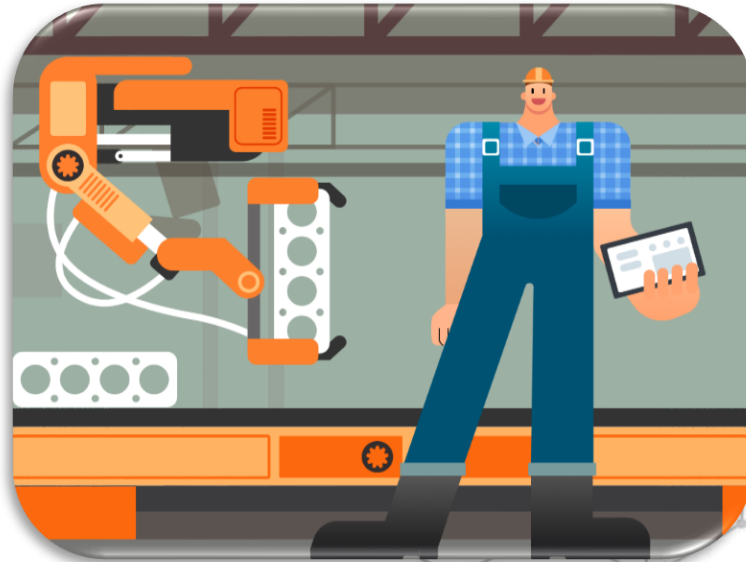
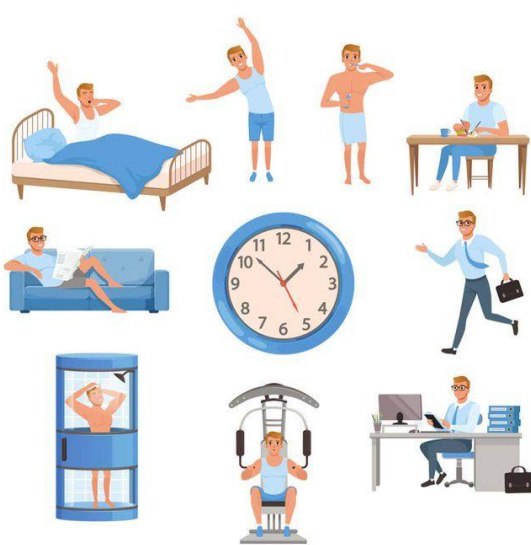
Medical Sources

- By one estimate, medical images consume up to one-third of global data storage (Intel white paper)



Internet of Things (IoT)

IoT data is measured in zettabytes, a unit equal to one trillion gigabytes. Cisco estimates that by the end of 2019, the IoT will generate more than 500 zettabytes per year in data—and in the years beyond, data will grow exponentially.



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Wearable Computing



Social Media

According to Domo's Data Never Sleeps 5.0 report, every minute of the day:

- Snapchat users share 527,760 photos
- More than 120 professionals join LinkedIn
- Users watch 4,146,600 YouTube videos
- 456,000 tweets are sent on Twitter
- Instagram users post 46,740 photos

More than a quarter (1.5billion) of the world's 7 billion humans are active on Facebook!

- There are five new Facebook profiles created every second!
- More than 300 million photos get uploaded per day
- Every minute there are 510,000 comments and 293,000 statuses updated
- Instagram has shown impressive growth even in Facebook is bigger.



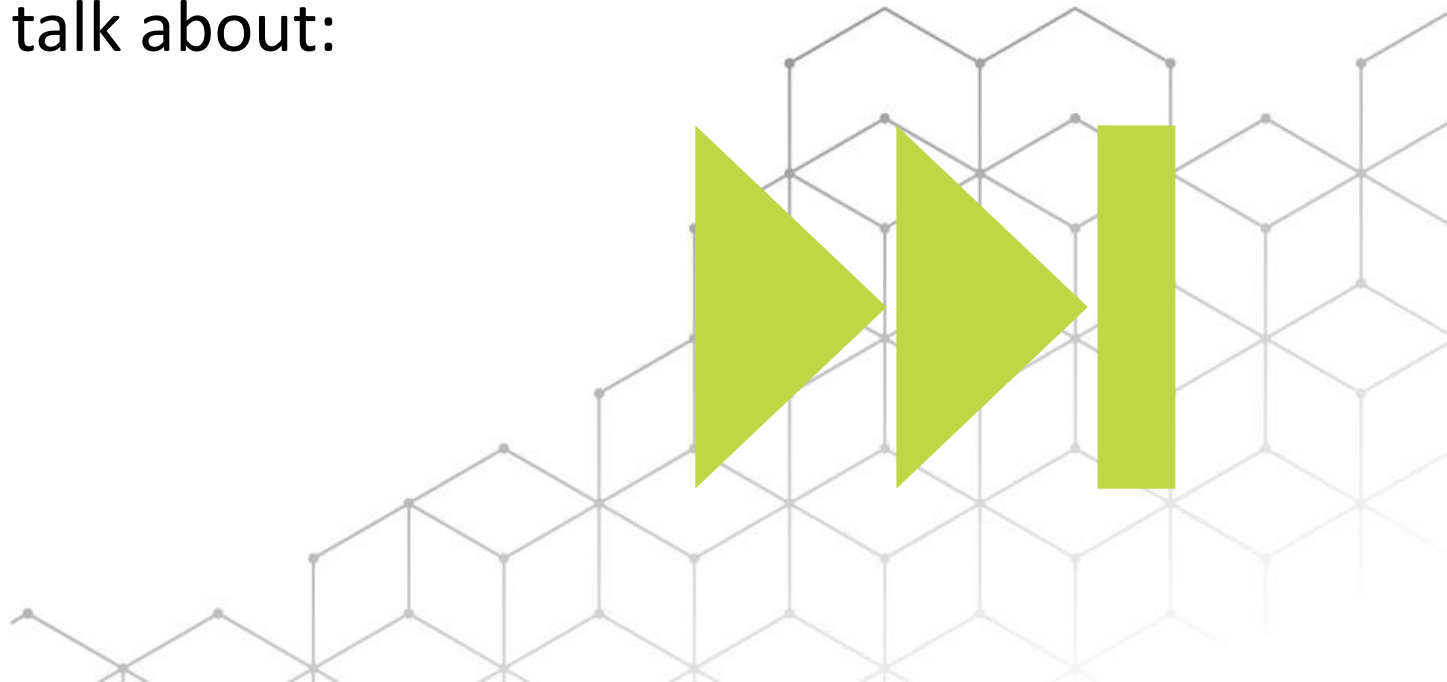
Warning

- Ilya Sutskever, co-founder of OpenAI, warns that the data available for training AI models is limited and compares it to fossil fuels that will eventually be depleted, while computing power continues to increase.



Next Steps...

- In the next theoretical lecture, we will talk about:
 - Big Data Platforms
 - Hadoop
 - Spark & Databricks
- In the Lab tomorrow, we will talk about:
 - Python Recall Session:
 - The Basics
 - The Lambda Functions
 - Error Handling
 - Advanced Data Structures
 - Hands-on Exercises



Questions?

The End

